What Can Money Buy?

Inequality and Fiscal Policy Implications of Citizens United v. Federal Election Commission

Alessio Rebechi[1] · Philippe Van Kerm[2,3] · Piotr Paradowski[3] · Anthony Lepinteur[2] · Nicholas Rohde[4]

[1]University of Tasmania · [2]University of Luxembourg · [3]Luxembourg Income Study · [4]Griffith University

IARIW General Conference · Brussels · 28 August 2026

Session: Recent laws and their inequality implications

Presented by the discussant: **Max Ghenis** (PolicyEngine)

Author response: Philippe Van Kerm

Does money in politics change actual policy?

The political effects of Citizens United are documented; the policy effects are not

In the last presidential election, the 100 largest donors gave **$2.4 billion** (OpenSecrets).

Citizens United v. FEC (January 2010) and **SpeechNow.org v. FEC** (March 2010) removed limits on independent political spending by corporations, unions, and individuals.

A decade of research finds **political** effects: higher Republican vote shares, more conservative state legislatures (Abdul-Razzak et al. 2020; Klumpp et al. 2016; Harvey & Mattia 2022).

> “What this implies for policy decisions, however, is unclear. Most research to date has found no impact on actual policy.”
>
> This paper supplies the missing link: from campaign finance deregulation to tax schedules to inequality.

Sixty years of limits, undone in two rulings

US campaign finance regulation, 1947–2022 (paper Figure 1)

1947	**Taft–Hartley Act**	Union contributions banned
1971–75	**FECA + FEC created**	Contribution and spending limits
1976	**Buckley v. Valeo**	Spending is speech; contribution limits stand
1990	**Austin v. Michigan**	States may limit corporate independent expenditures
2002	**BCRA (McCain–Feingold)**	Soft money and electioneering restricted
Jan 2010	**Citizens United v. FEC**	Corporate and union outside spending unlimited; state bans invalidated
Mar 2010	**SpeechNow.org v. FEC**	Unlimited contributions to independent-expenditure groups: Super PACs
2014	**McCutcheon v. FEC**	Aggregate contribution limits struck down
2022	**FEC v. Ted Cruz**	Post-election contribution limits struck down

The natural experiment: 23 states had their own bans

Citizens United erased all of them within months — differentially

Corporate *and* union ban — 13 states in sample

AK, AZ, MI, NH, NC, ND, OH, OK, PA, RI, TX, WI, WY

Corporate ban only — 8 states

CT, IA, KY, MA, MN, MT, TN, WV

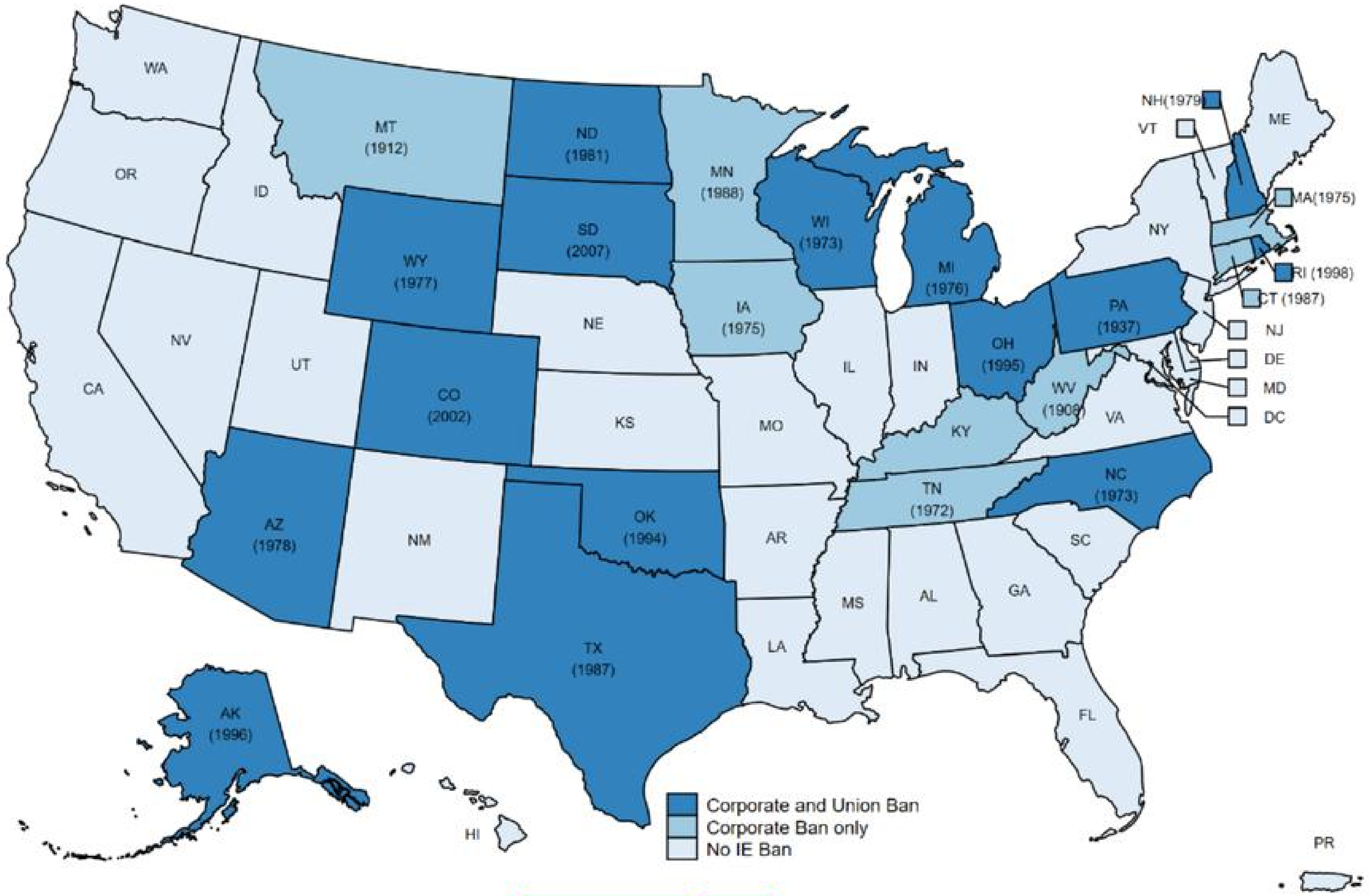
No ban — controls

The remaining states never restricted independent expenditures

---

All bans revised to comply by **November 2010**: a sharp, court-imposed, differential deregulation. CO and SD (bans adopted only in 2002 and 2007) are excluded.

Figure 2: State Campaign Regulations before Citizens United



Notes: The map has been reproduced from Klumpp et al. (2016). Years in parentheses indicate the years of introduction of the funding restrictions.

What followed: an explosion of outside spending

Outside spending by recipient party, $ millions (paper Table 1; OpenSecrets)

	2004	2024	Growth
For Democrats	60	1,062	**×18**
For Republicans	23	763	**×33**
Against Democrats	4	1,411	**×353**
Against Republicans	7	923	**×132**

Negative spending grew fastest — money spent *against* Democrats rose 353-fold over twenty years.

And it went dark: the share of Super PAC spending with **full donor disclosure** fell from 77% (2010) to 33% (2024) (paper Table 3).

This paper

From vote shares to tax schedules to inequality

RESEARCH QUESTION

Did lifting bans on independent political expenditures change the size and progressivity of state income taxes — and post-tax income inequality?

Closest prior work (Slattery, Tazhitdinova & Robinson 2023, *J. Public Economics*) studies statutory rates and aggregate revenues — the tax system *in the books*.

This paper measures the tax system **as households experience it**: average rates by income and wealth group, and the redistribution the schedule actually delivers.

> Preview: where states had banned **both corporate and union** spending, top-group tax rates and redistribution fell after the ruling. Where only corporations had been banned, they did not.

Measurement: 900 simulated tax systems, one fixed population

SCF microdata through NBER TAXSIM — only the law varies

Survey of Consumer Finances 2011 (LWS-harmonized): the household survey that oversamples the wealthy — credible top-5% and top-1% incomes *and net worth*.

NBER TAXSIM: federal and state income tax law for all 50 states, 2004–2021.

The same households are run through **every state-year schedule** (incomes CPI-deflated to each year):

50 states × 18 years = 900 datasets

Why this is clean

With the population held fixed, no composition change, migration, or behavioral response contaminates the measure. What moves is **tax legislation itself** — a pure *de jure* policy panel.

Main text: state taxes assessed on pre-federal income (AGI). Appendix D repeats everything on after-federal-tax income — results are larger and sharper there.

(The SCF is triennial: "2011" is the 2010 wave, LWS US10, with incomes for 2009 — see the replication slide.)

Summarizing schedules, not statutory parameters

Top marginal rates miss deductions, exemptions, and the base — these indicators capture the whole shape

Indicator	Definition	Captures
Average tax rate (ATR)	Total state income tax over total pre-tax income	Size of the tax
ATR, top 5% and top 1%	By income — and, using SCF strengths, by net worth	Burden at the top
Progressivity (β)	ATR of top 20% minus ATR of bottom 20%	Shape of the schedule
Reynolds–Smolensky index	Gini(pre-tax) − Gini(post-tax)	Redistribution delivered

Each indicator Φ is evaluated on the fixed household sample for every state-year law — turning 900 simulated tax systems into a 900-cell panel of schedule characteristics.

Identification: three complementary designs

46 states, 2004–2021, N = 846 · treated = pre-2010 ban states · controls = never banned

$$\text{Outcome}_{st} = \beta_1(CU_t \times \text{CorpBan}_s) + \beta_2(CU_t \times \text{CorpUnionBan}_s) + \gamma_s + \delta_t + \varepsilon_{st}$$

Two-way fixed effects

Separate treatment effects by pre-existing ban type — the paper's key refinement over pooled designs.

Event studies

Leads and lags around 2010 (2009 omitted): pre-trend tests plus the time path of effects.

Synthetic DiD

Arkhangelsky et al. (2021) reweighting, group-level and state-by-state ATETs; plus a Wooldridge (2025) TWFE check.

Exclusions: CO and SD (bans adopted only in 2002 and 2007), NE (nonpartisan unicameral legislature), LA (nonpartisan blanket primaries). Standard errors clustered by state; coefficients ×100 (percentage points).

First stage confirmed: legislatures moved right

Replicates Abdul-Razzak et al. (2020) and extends it through the 2022 elections (paper Appendix A)

Republican vote share: ≈ +4–5pp in ban states after 2010, both chambers (significant with trend controls).

Median legislator ideology (Shor–McCarty): +0.20 to +0.41 toward conservative.

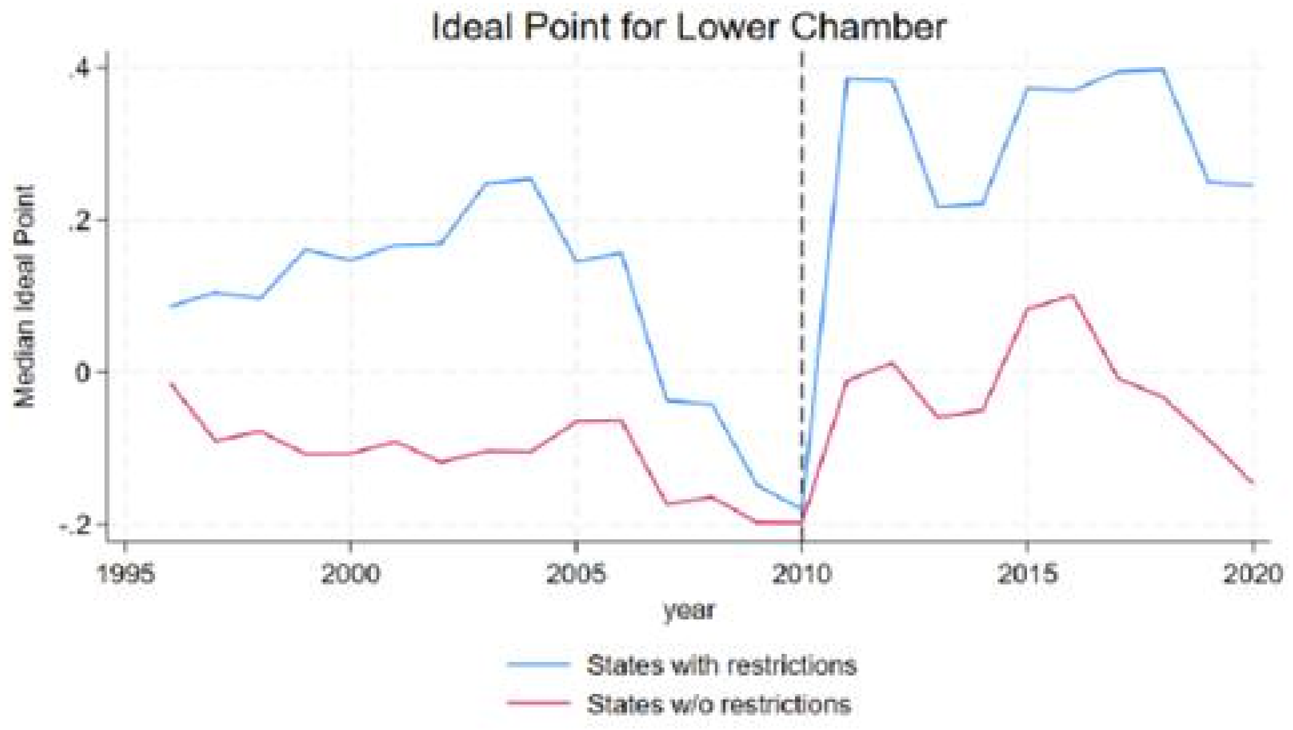
Party control: Democratic legislative control −21pp**; Republican trifectas +15pp*.

Majority parties became more extreme; minority parties moderated. Polarization itself moved little.

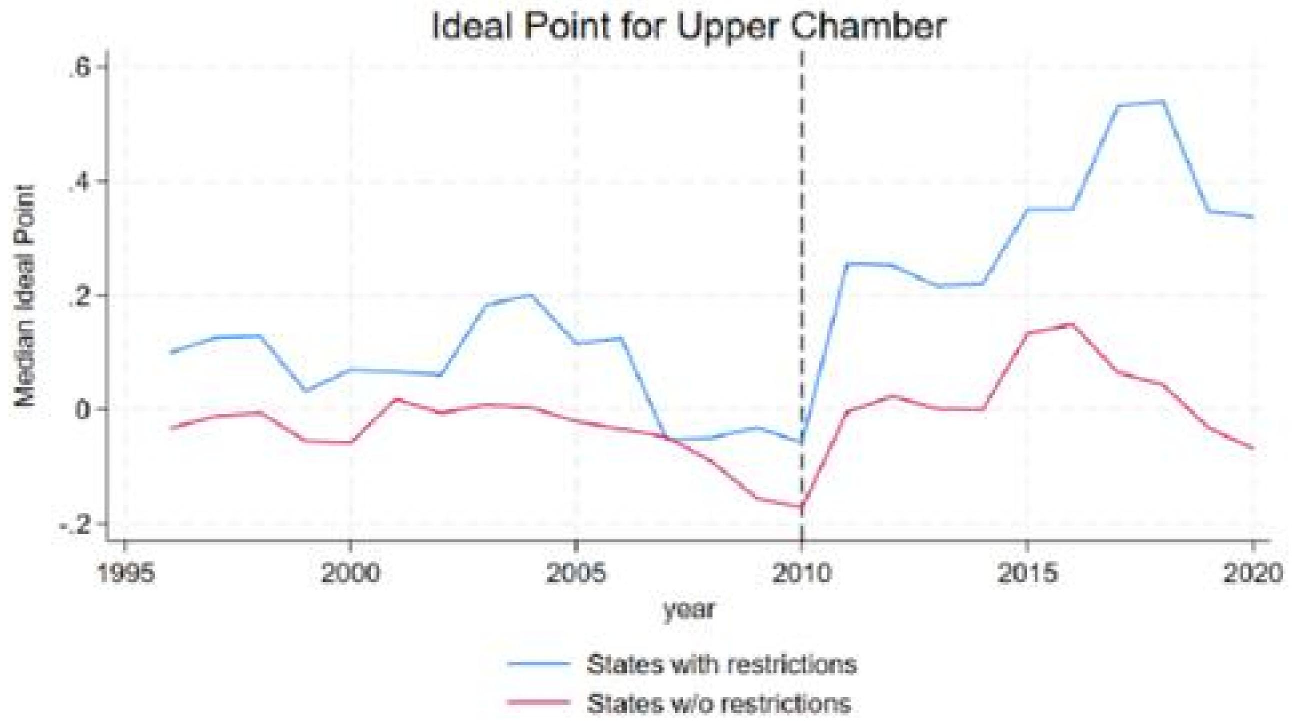
> By ban type (Table A-4): vote-share effects are not significant in either group, and the ideology shift is significant only in corporate-only-ban states (+0.27**).

Figure A–3: Trends in the Median Ideology by Chamber

(a) Lower Chamber



(b) Upper Chamber



Main result: top rates and redistribution fell — only where both bans fell

DiD estimates × 100 (percentage points), state income taxes, 2004–2021 (paper Table 4)

Citizens United ×	ATR	Top 5%	Top 1%	Top 5% (wealth)	Top 1% (wealth)	Reynolds–Smolensky
Corporate ban only	0.01	0.14	0.30	0.07	0.10	0.07
Corporate & union ban	−0.01	−0.33	**−0.53****	−0.25	**−0.36***	**−0.11****
Difference (both − corp. only)	−0.02	**−0.47****	**−0.83****	**−0.32***	**−0.46***	**−0.18****

Concentrated at the top. The overall ATR barely moves; top-1% rates fall 0.53pp.

Asymmetric. Corporate-only-ban states drift (insignificantly) the other way.

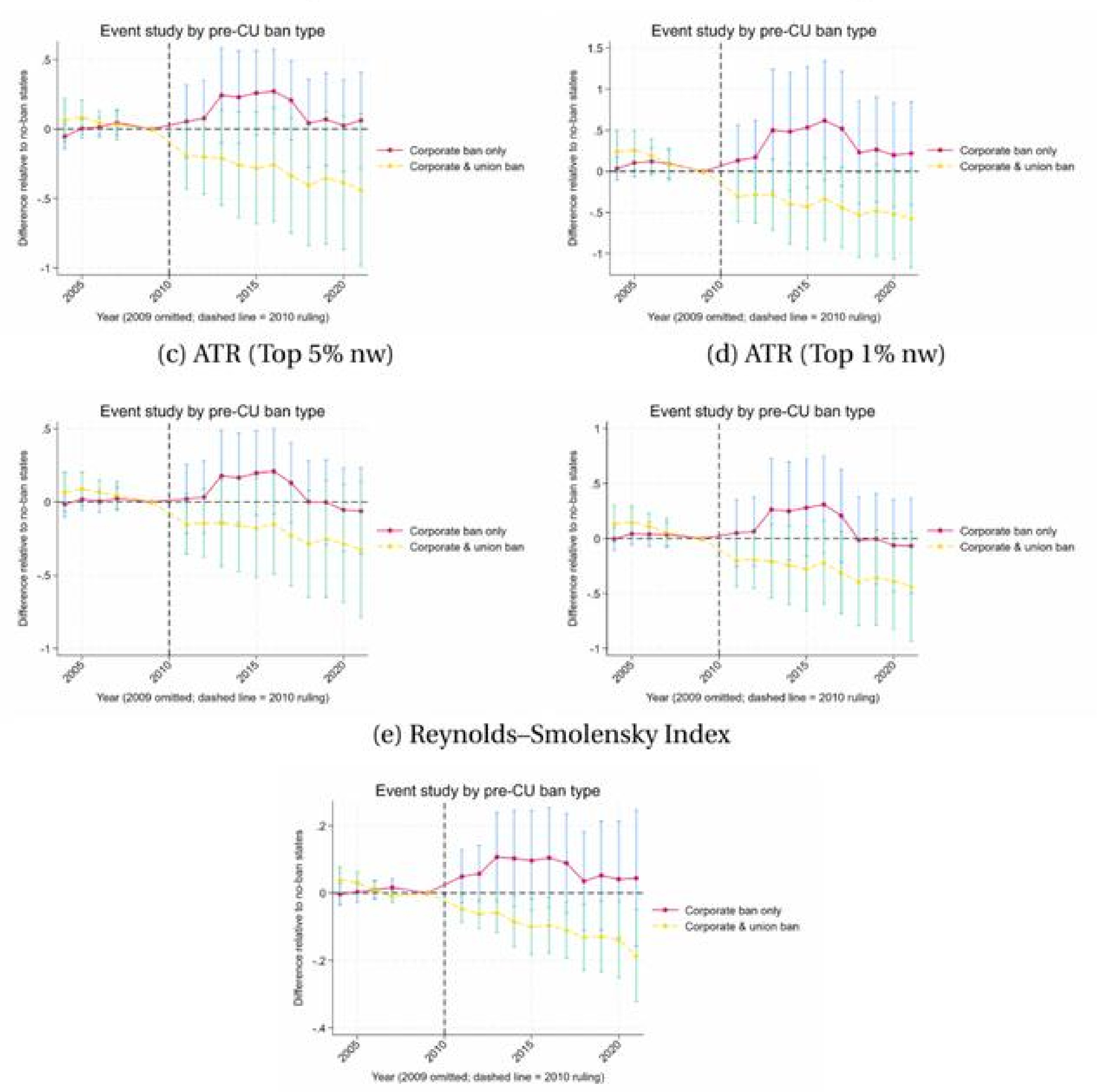
Redistribution falls. Reynolds–Smolensky drops where both bans lifted.

N = 846; SEs clustered by state; * $p<0.10$, ** $p<0.05$.

No significant pre-trends; gradual divergence after 2010

Event studies by pre-existing ban type (paper Figure 3) — joint pre-trend tests pass for every outcome

Figure 3: Event Study Graphs for State-Level Taxes by Ban Type (State Taxes only)



Notes: Each panel reports coefficients from an event-study regression that interacts treatment status by ban type with year-relative-to-2010 indicators, using 2009 as the omitted category. State and year fixed effects are included.

Corporate-and-union-ban states (yellow) drift steadily down; the gap builds over a decade rather than jumping.

Post-2010 coefficients jointly significant for that group: top 1% p = 0.025, Reynolds–Smolensky p = 0.049 (paper Table 5).

Ban-type differences jointly significant for every outcome except the overall ATR.

The same signs under stress — with two caveats

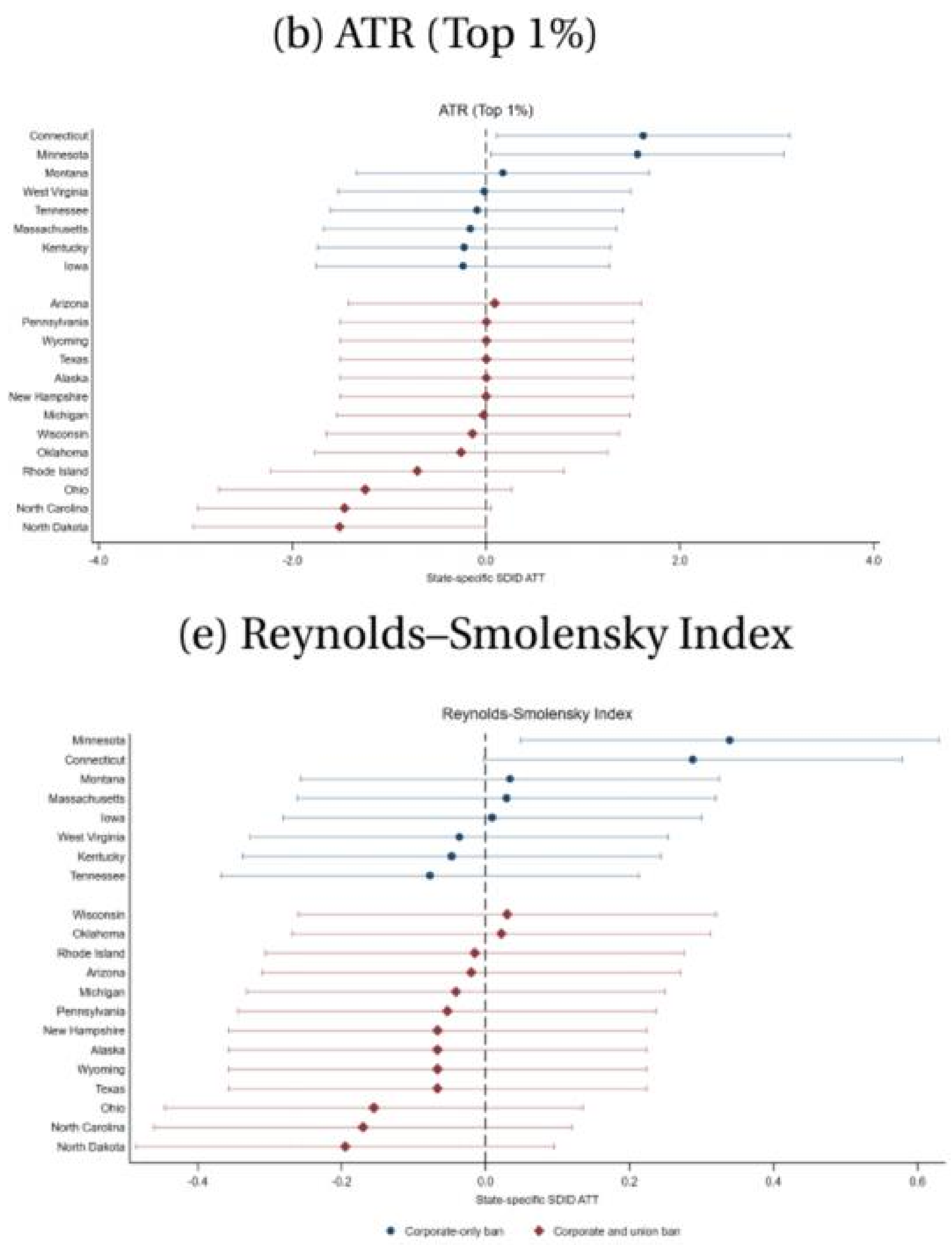
Alternative estimators, samples, and income concepts

Synthetic DiD	Same signs; group estimates smaller and not significant (top 1% −0.40, RS −0.07); ban-type differences hold at 10% (−0.72∗, −0.13∗)
Wooldridge (2025) TWFE	Identical to baseline (top 1% −0.53∗∗, RS −0.11∗∗) — expected with one treatment date and never-treated controls
Leave-one-out	No single treated state drives either group
Drop 8 no-income-tax states	Effects grow (difference: top 1% −1.13∗∗∗, RS −0.20∗∗) — baseline is a lower bound
After-federal-tax income (App. D)	Larger and sharper (top 1% −0.93∗∗∗; difference −1.27∗∗∗, RS −0.24∗∗∗)

The honest caveat: adding state-specific linear trends pulls the corporate-and-union effects "much closer to zero" (Fig. 4), and the SDID group estimates lose significance. The authors treat trends as a robustness check, since they can absorb genuinely gradual treatment effects — exactly the dynamics the event studies show.

Where the effect lives

State-by-state synthetic DiD effects (paper Figure 5): top-1% ATR and Reynolds–Smolensky



Connecticut and Minnesota pull the corporate-only group positive — states mostly under Democratic or divided government.

North Carolina, North Dakota, Ohio anchor the negative end — Republican-controlled after 2010.

> The authors say it plainly: "most of our results are driven by few outlying states" — though leave-one-out shows no *single* state is decisive.

Channels: politics predicts taxes — with a residual

Political outcomes as transmission mechanisms (paper Tables 7, 8, C-2)

Political composition moves tax outcomes

Republican Senate vote share → redistribution down (RS −0.14∗). More conservative median ideology → top-1% ATR −0.29∗∗. Democratic trifectas and governors → higher top rates and more redistribution; Republican control → the reverse.

...but does not fully explain the ruling's effect

Residualize each tax outcome on vote share, ideology, or polarization, then re-run the DiD: corporate-and-union effects persist (top 1% ≈ −0.5pp, RS ≈ −0.1, still significant).

Reading: electoral composition carries part of the effect. The remainder points to influence that does not show up in seat counts — **lobbying, donor pressure, agenda-setting**.

Why the asymmetry? The union counterweight

Union density by pre-existing ban type (paper Figure 6)

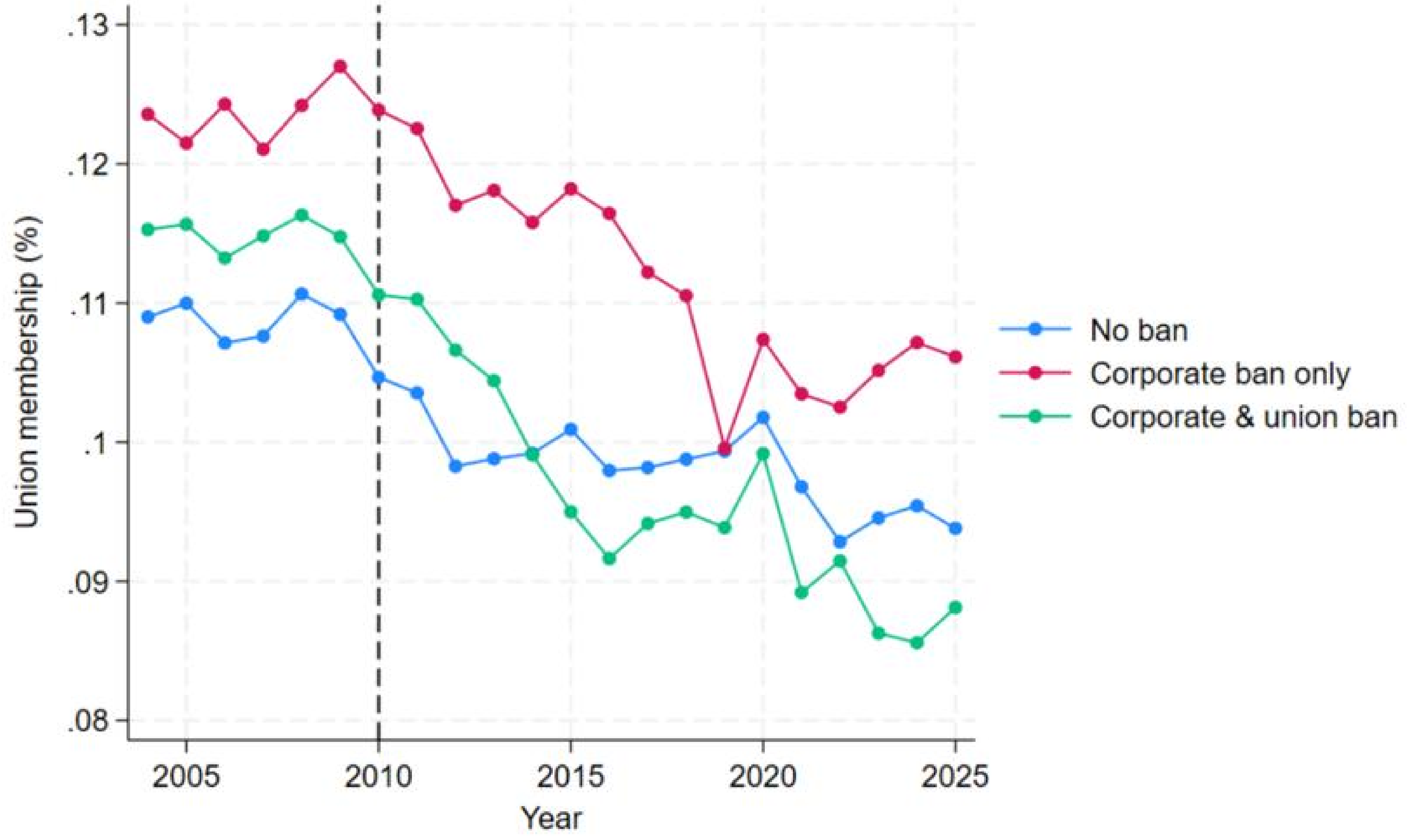
Unions send ≈ **87%** of their contributions to Democrats; corporate money leans Republican (2023–24 cycle, OpenSecrets).

Corporate-only-ban states: unions could already spend — and are strongest there. New corporate money met an existing counterweight, so competition stayed roughly balanced.

Corporate-and-union-ban states: historically Republican-leaning, weaker unions. Lifting both bans favored the side with deeper pockets — the regressive tilt.

In the authors' words: the ruling "amplified pre-existing asymmetries in political influence."

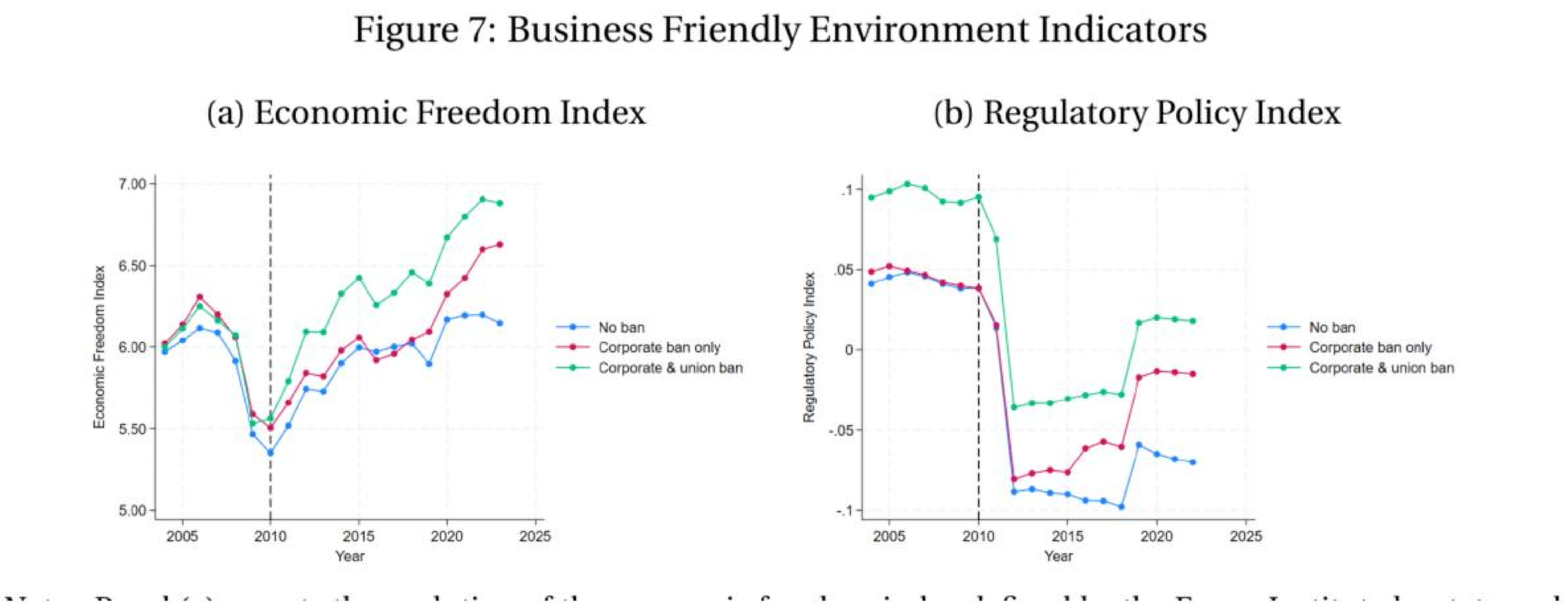
Figure 6: Union Density by Ban Type



Notes: The figure reports average union membership rates, defined as the percentage of employed workers who are union members, computed at the state-year level (Hirsch et al. 2026) and grouped by pre-existing campaign finance regimes across all sectors.

Part of a broader market-oriented drift

Economic freedom (Fraser) and regulatory policy (Cato) indices by ban type (paper Figure 7)



After 2010, both indices rise most in the states that had banned both corporate and union spending — descriptive, but consistent with the tax findings: a policy environment tilting toward business interests, not just a one-off rate cut.

The authors' conclusions

Citizens United changed not only who legislates, but what they legislate

1. Where both corporate and union bans fell, state income taxes became **persistently less progressive**: lower top-group rates, less redistribution, higher post-tax inequality.

2. Effects unfolded **gradually** — policy responds over a decade, not an election cycle.

3. Transmission runs partly through **electoral composition**, partly through influence beyond it.

> "Wealthy individuals and corporations can influence policy agendas by financing campaigns and supporting organizations that promote market-oriented and conservative ideologies, resulting in fiscal policies that disproportionately favor their interests ('wingnut welfare')."

Discussion

Four comments and a public-data replication — from a tax-microsimulation seat

1 · The finding is about the union ban — say so up front

Every significant coefficient sits in the 13 states where the *union* ban was also lifted; the abstract describes a general effect

Table 4, read as a triple difference

Corporate-&-union ban: top 1% **−0.53∗∗**, RS **−0.11∗∗**. Corporate-only: +0.30, +0.07 (n.s.). The −0.83∗∗ difference is, under DiD logic, *the extra effect of also lifting the union ban* — the opposite of what a unions-fund-Democrats story predicts.

The detectable political first stage is in the corporate-only group (Table A-4)

By ban type, Republican vote share is n.s. in both groups (0.05, 0.01); the ideology shift is significant only in the *corporate-only* states (+0.27∗∗) — where taxes moved up — and the cross-group differences are n.s. Residualizing on vote share or ideology barely moves the tax coefficients (Table 8: −0.47 to −0.56).

The authors' resolution — corporate-&-union states had weaker unions — rests on a ~1pp density gap (Fig. 6). In our public-data replication pre-2010 density is 11.5% vs 12.4% vs 11.5% across the three groups.

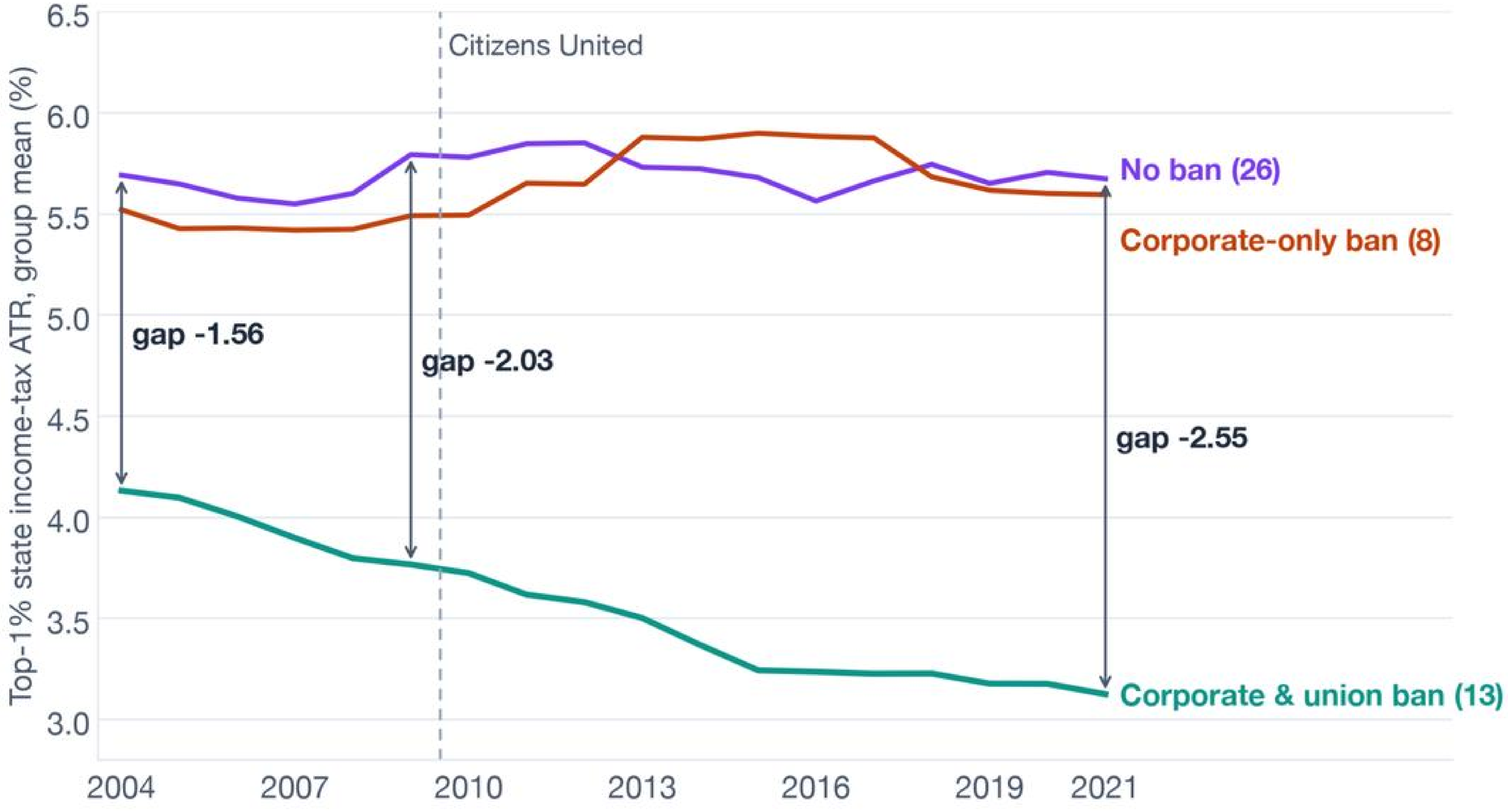
What would convince me

- State the conditionality in the abstract and title the mechanism section after it.
- A dose-response test: CU × pre-2010 union density, within ban groups. In our replication the corporate-&-union effect does *not* scale with density (interaction −0.02, p = 0.57), and CU × density is *positive* — high-union states raised top taxes after 2010.
- We ran the obvious horse race: adding post × Republican-trifecta acquisition (2011–13; 8 of your 13 states, 1 of 8 corporate-only, 6 of 26 controls) moves the coefficient only 16–20% (top 1% −0.57∗, RS −0.07∗); REDMAP targeting, ≈ 0. So the contrast is not just the 2010 wave — which leaves the question sharper: what is it? A theory of change with names would answer it.

Replication: SCF 2010 public extract → TAXSIM-35 (local), 47 units × 18 years; union density from Hirsch–Macpherson (unionstats.com). Everything is open: github.com/MaxGhenis/iariw-2026-discussant-slides

2 · Which −0.53 do you believe?

The treated–control gap was already widening before 2010 — replication from public data (SCF 2010 → TAXSIM-35, same design), group means by year



Gap −1.56 (2004) → −2.03 (2009) → −2.55 (2021): the pre-period slope is at least as steep as the post-period slope. Joint pre-test p = 0.40 only because clustered SEs are wide.

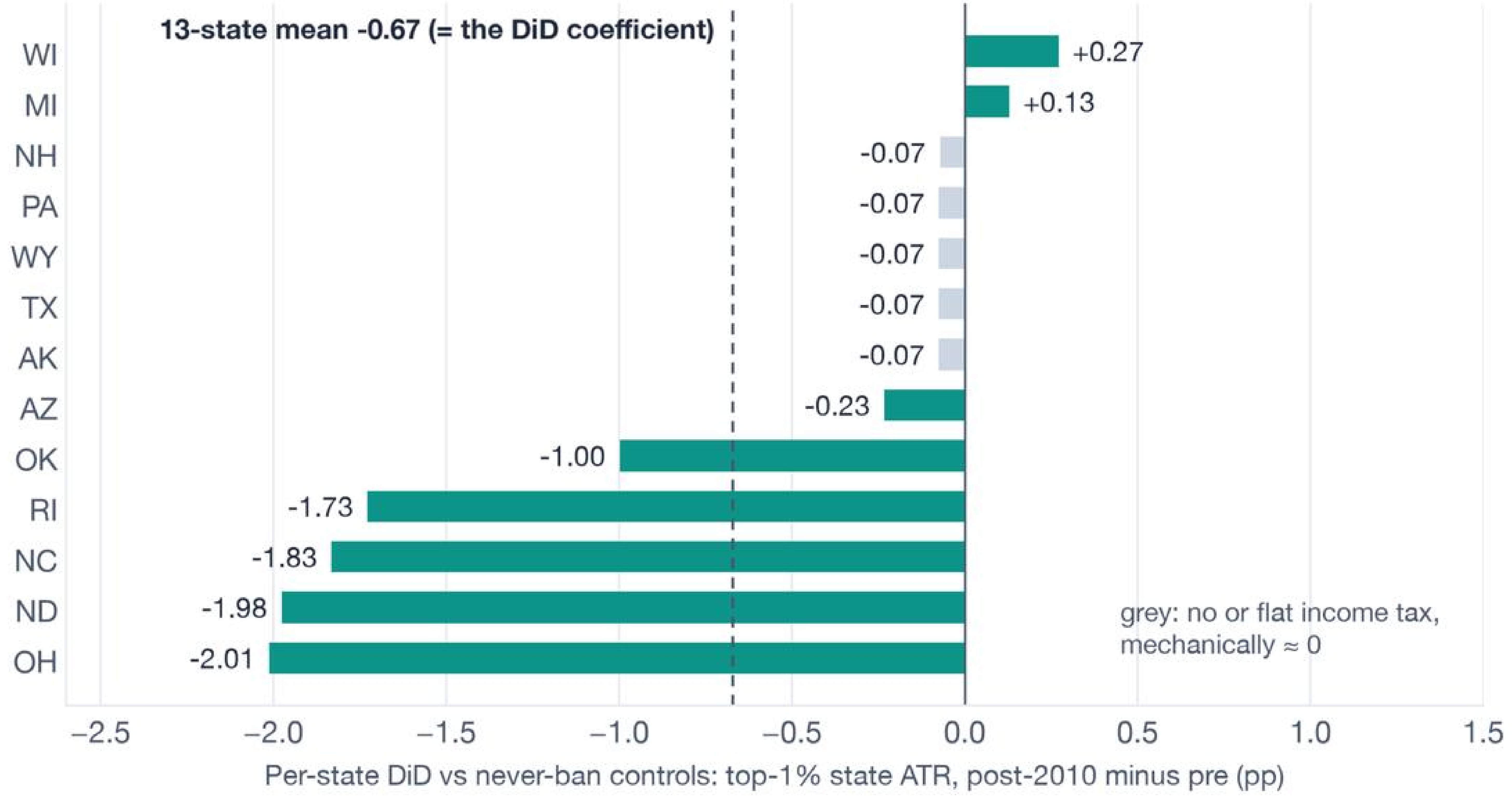
Consistent with the paper's own checks: state-specific trends pull the effect "much closer to zero" (Fig. 4; −0.27, n.s., in our panel); SDID group estimates are n.s. (Table 6: −0.40, −0.07).

> **Asks**
> Report the trends specification as co-equal, and bound the estimate with Rambachan–Roth pre-trend sensitivity.

In fairness: the stars are not a few-cluster artifact. In our replication the top-1% effect survives randomization inference (p = 0.013) and a wild cluster bootstrap (p = 0.030).

3 · Five tax reforms, not thirteen states

Per-state contributions to the corporate-&-union coefficient (replication; the paper's Fig. 5 shows the same ranking with SDID)



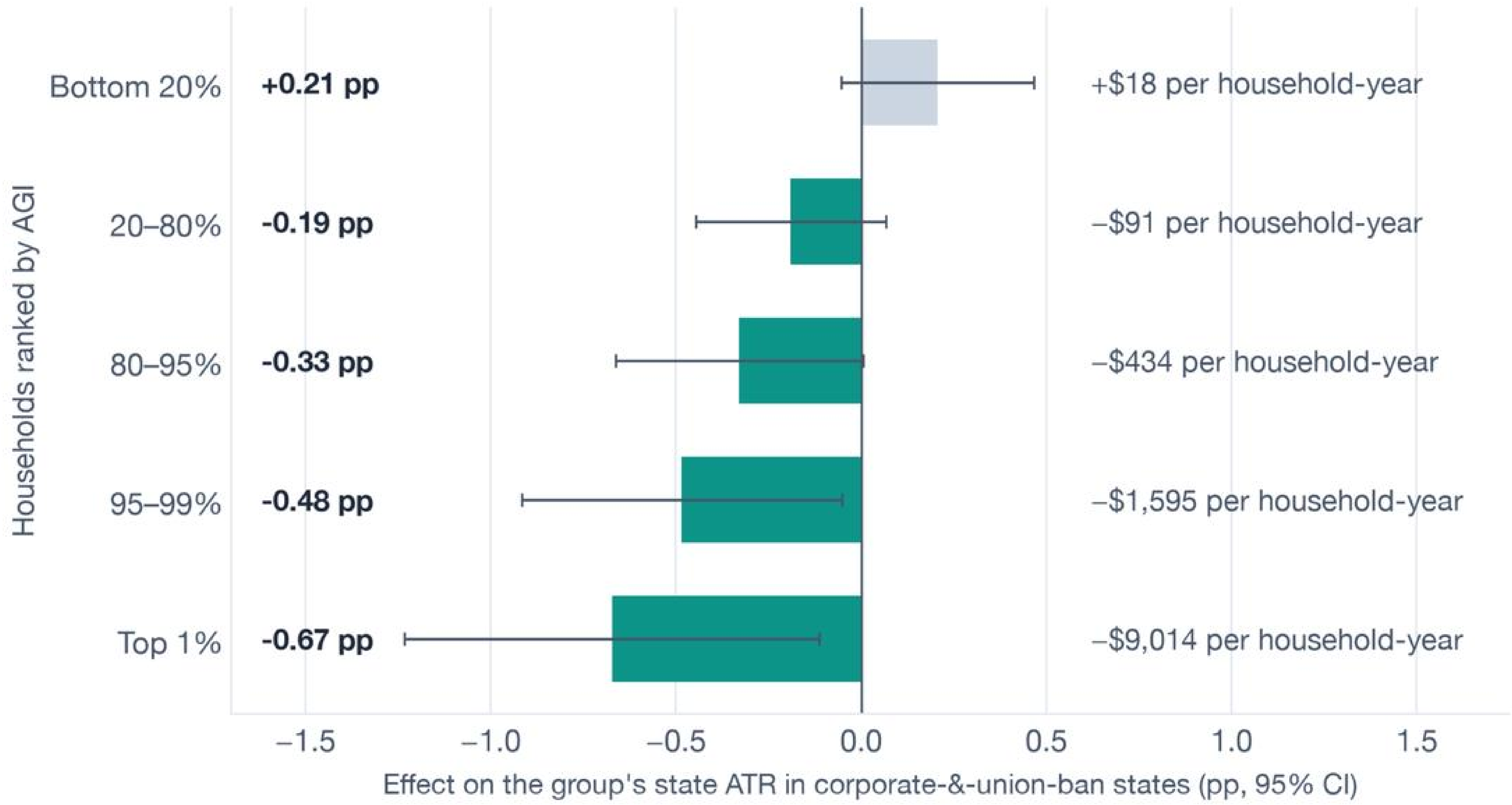
Ohio, North Dakota, North Carolina, Rhode Island and Oklahoma carry the estimate. Four treated states have no broad income tax and Pennsylvania's is flat — they cannot respond. Michigan and Wisconsin moved the other way.

Drop NC, ND and OH together: **−0.29 (SE 0.24, p = 0.22)**. Leave-one-out alone cannot show this.

> **Asks**
> Consider making the income-tax-states sample (Table B-1) the main one; report leave-three-out; and tell the story of the five reforms — North Dakota's cuts began in May 2009, during the oil boom: a plausible confounder, not a channel.

4 · Model the tax function, not six summaries of it

The same 900 simulated datasets support a household-level incidence regression (here in its group-level equivalent) — who got the cut, in dollars (replication, corporate-&-union states)



A monotone gradient: nothing at the bottom, about −$9,000 a year for the average top-1% household (2010 dollars). Capital-heavy and wage-heavy top households are hit alike — across-the-board top-rate cuts, not capital carve-outs.

With a fixed sample, the saturated micro regression *is* the paper's Φ by subgroup — so this costs nothing new and turns "RS −0.11" into an incidence table. And contrasts difference out common schedule shocks: the β the paper defines in eq. (5) but never reports — ATR(top 20%) − ATR(bottom 20%) — is its most precise outcome in our replication: −0.68 (SE 0.22, t = 3.1).

Asks
Report β (eq. 5, defined but never shown) and the dollar incidence; say that RS −0.11 is −0.0011 Gini points on a state-tax redistribution of about 0.004 — and say whether benefits B (eq. 1) enter the simulation at all.

The headline reproduces from public data

SCF 2010 public extract → NBER TAXSIM-35 (run locally) → 51 jurisdictions × 18 years × 6,482 households → eq. (7). Coefficients × 100, SEs clustered by state

Table 4 row	Paper	Replication
Top-1% ATR, corporate & union ban	−0.53** (0.24)	**−0.67** (0.29)**
Reynolds–Smolensky, corporate & union ban	−0.11** (0.04)	**−0.09** (0.03)**
Difference (both − corporate only), top 1%	−0.83** (0.35)	**−0.87** (0.38)**
Corporate-only arm	positive, n.s.	**positive, n.s.**
Income-tax states only (Table B-1), top 1%	−0.77** (0.30)	**−0.95** (0.36)**
Overall ATR, corporate & union ban	−0.01 (0.13)	**−0.35** (0.17)**

N = 846 only with DC in the panel (47 × 18) — the text says 46 states. "SCF 2011" is the 2010 wave (LWS US10; income year 2009): the SCF is triennial.

What does not reproduce: the clean zero on the overall ATR. The public extract forces no itemized deductions and taxes Social Security as pensions, which inflates middle-class liabilities. The top-1% and RS rows are insensitive to this.

Full pipeline ≈ 25 minutes of compute; one SCF implicate; TAXSIM vintage differs. Scripts, panel and results: github.com/MaxGhenis/iariw-2026-discussant-slides. A released 47 × 18 panel plus code would make the core table reproducible in an afternoon.

Where this sits in the Citizens United evidence

What the cited and uncited empirical work actually finds — every entry retrieved and checked

Gilens, Patterson & Haines (2021, APSR) — not cited	Same natural experiment, generalized synthetic control: top corporate tax −2.83 ($p = 0.01$; 0–100 index), and twice as large in the eight corporate-only states (−5.55, vs all treated) — the union counterweight hypothesized ex ante, in the mirror image of this paper.
Slattery, Tazhitdinova & Robinson (2023, JPubE) — cited as "a moderate effect"	A precise null on top corporate, top personal-income and sales rates, incentives and revenues (top PIT ≈ −0.46pp, n.s.); corporate-only restriction: a tighter null. Parallel trends fail in levels, so they use logs.
Klumpp, Mialon & Williams (2016, JLE)	CU raised Republican House-race wins ≈ 4pp; descriptively largest in MN, MT, MI, OH, IA — three of them corporate-only states (no ban-type test reported). The corporate-funded 2010 REDMAP drive targeted MI, OH, PA, TX, NC, WI for redistricting, not for ban type.
Abdul-Razzak, Prato & Wolton (2020, Electoral Studies)	Vote-share effect ≈ 4–5pp (trend-dependent); ≈ 8pp where unions are weak, ≈ 0 where strong — by density, not ban type. In treated disclosure states Democratic-aligned groups still outspent Republican-aligned ones after CU; fn. 12: no effect on public goods.
Hansen, Rocca & Ortiz (2015, JOP); Spencer & Wood (2014)	Major corporations "were not a source of the dramatic increase in independent spending" (federal, 2012); the state-level surge ran through 501(c)/527 vehicles whose donors are unobservable — ≈ $1.6M per treated state over 2010–12.
Akey et al. (2023, NBER); Werner & Coleman (2015, JLEO); Farver (2024)	The policy record is mixed, not "mostly null": enforcement and antitakeover effects exist; tax rates negative but n.s. (PIT ≈ −0.3pp); no effect on environmental policy or overall policy liberalism; turnover rose in both parties.

Questions to open the floor

Federal analogue. There is no US-without-Citizens-United counterfactual at the federal level. Do these state results bound what happened in Congress?

The 2021–2023 tax-cut wave. Many corporate-and-union-ban states cut income taxes after the sample ends in 2021. Does extending the panel sharpen the effect?

Who actually spent? Your Table 1 shows the two sides near parity by 2020–24 at the federal level. What does the state-level ledger look like in the 13 states that drive the result?

Thank you — and over to **Philippe Van Kerm**

Max Ghenis · max@policyengine.org · slides, memo, replication: github.com/MaxGhenis/iariw-2026-discussant-slides